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Prevention of Turbocharger Damage After Engine Mechanical Issue

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Introduction

This document addresses the possibility of turbocharger damage after engine mechanical issues. Engine mechanical issues that can put debris into the air handling system (including the exhaust system, exhaust gas recirculation (EGR) system, intake system, and the air intake heater) are rare. They could include piston, intake valve, and exhaust valve issues. However, if **not** properly repaired, these issues can lead to subsequent multiple turbocharger replacements, due to damage from debris that was **not** removed from the air handling system.

Verification of Issue

⚠ CAUTION ⚠

All debris must be removed from the air handling system (including the exhaust system, EGR system, intake system, and the intake air heater) after an engine mechanical issue. Failure to remove all debris can lead to turbocharger damage and subsequent multiple turbocharger replacements.

⚠ CAUTION ⚠

Appropriate engine model Clean Care Kit must be used to prevent debris from entering the air intake system.

Any of the listed engine types listed below that have had a piston, intake valve, or exhaust valve mechanical issue:

- All engines with exhaust gas recirculation (EGR) and a variable geometry turbocharger (VGT).

Symptoms can include those normally present for mechanical issues, such as:

- Noise
- Vibration
- Running rough.

Also, look for multiple turbocharger replacements with Fault Code 2449, Fault Code 1898, Fault Code 3616, or Fault Code 2387, and stuck sector gears at very short intervals.

Debris particles as small as 0.70 mm [0.030 in] caught in the air handling system are difficult to clean out. This debris can travel through the EGR system, back into the intake, and out the exhaust manifold to the turbocharger, where it causes damage that prevents motion of the variable geometry turbocharger (VGT) mechanism.

Resolution of Issue

If an engine experiences a piston, intake valve, or exhaust valve mechanical issue, it is important to understand that the small size of the debris particles could possibly allow them to travel through the air handling system and become lodged in various components (See Figure 1).

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Figure 1 - Example of exhaust valve debris

Note : Some debris pieces are large, however, particles as small as 0.70 mm [0.030 in] could damage a turbocharger.

Follow the normal procedures for repair of these types of issues, along with the following additional steps. See corresponding Service Manual. Reference procedure 200-004 in Section F for the air intake system component location on the engine. Reference procedure 200-005 in Section F for the exhaust system component location on the engine.

⚠ WARNING ⚠

When running the engine with turbocharger not installed, keep hands and face away from exhaust manifold. Wear appropriate eye, face, and hand protection. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

1. Replace the EGR cooler.
2. Inspect the interior of the following components. Flush the air passages in these components using safety solvent. Rinse and dry with compressed air. See corresponding Service Manual. Reference procedure 200-004 in Section F for the air intake system component location on the engine. Reference procedure 200-005 in Section F for the exhaust system component location on the engine.
 - a. All EGR air plumbing
 - b. EGR measurement venturi (ISM and ISX engines)
 - c. Air intake connection (Including intake air heaters where applicable)
 - d. Intake manifold and cover
 - e. Exhaust manifold (for multiple piece manifolds, remove sections and clean)
 - f. Intake and exhaust passages in the cylinder head (if the cylinder head is to be reused)
 - g. EGR valve
 - h. Charge air cooler (CAC) **must** be cleaned per original equipment manufacturer (OEM) guidelines.
 - i. Intake air throttle (if applicable)
3. Operate the engine without the turbocharger installed at low idle for approximately 3-5 minutes to be sure the system is free of debris.

Note : The following lines need to be blocked off in order to operate the engine without the turbocharger.

- a. Turbocharger coolant supply
- b. Turbocharger coolant return

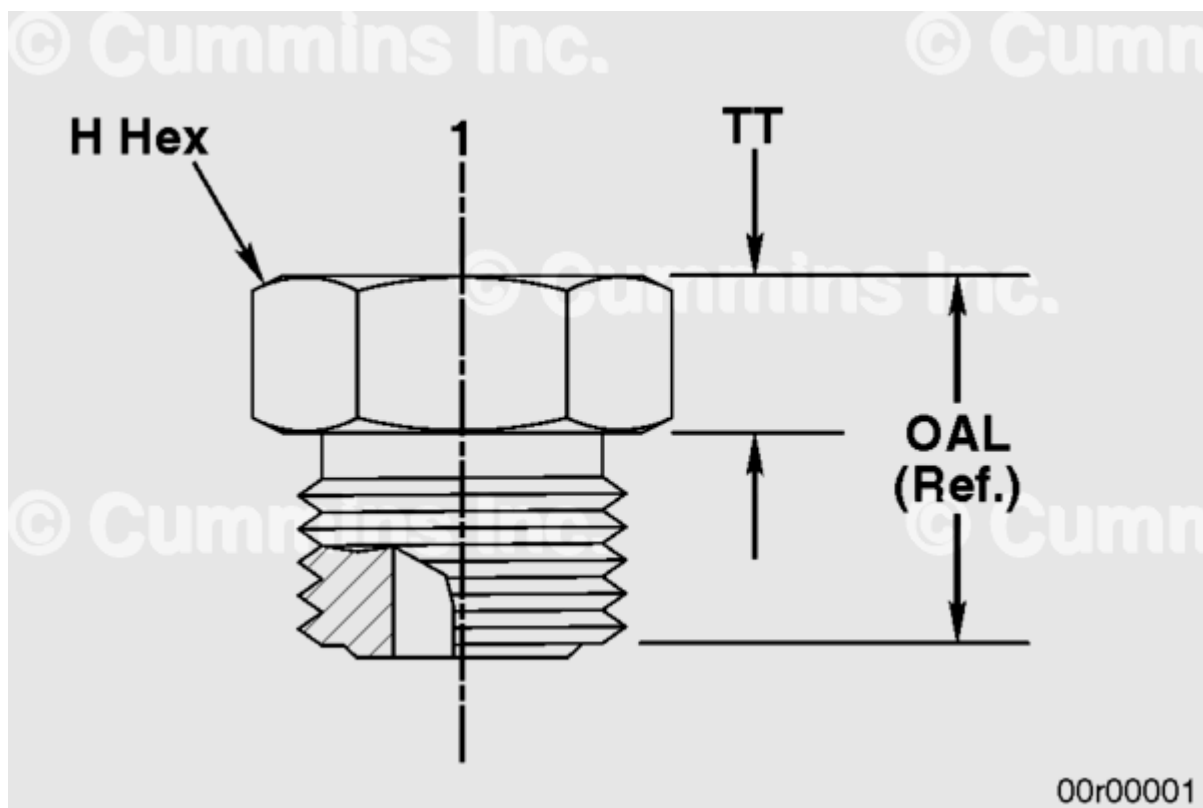
- c. Turbocharger actuator coolant return
- d. Aftertreatment fuel injector coolant supply (if applicable)
- e. Turbocharger oil supply.

Note : Due to the different engine models listed, exact quantities for the plugs and caps are not listed in Table 1 and Table 2. Please refer to the specific engine to determine the type and required number of plugs (Table 1) and caps (Table 2). Actual sizes for the caps and plugs are listed if the Parker™ part is not available. (See Appendix A)

4. After Step 1, 2, and 3 are completed and the engine is assembled with the turbocharger installed, operate the engine at low idle for 2 minutes while deaerating the cooling system. Use low idle for this procedure, to keep the turbocharger speed low while any residual debris is blown out of the system.

Appendix A

Table 1 lists the required plug sizes for blocking off oil and/or coolant.



O-Ring Face Seal Plug dimensions - See Table 1

Table 1 - O-Ring Face Seal Plug				
Parker™ Part Number	Dimension 1 (inch)	Dimension H (inch)	Dimension OAL (inch)	Dimension TT (inch)
6 PNLO	3/8	3/4	0.76	0.32
8 PNLO	1/2	7/8	0.86	0.35

Table 2 lists the required cap sizes for blocking off oil and/or coolant

O-Ring Face Seal Cap dimensions - See Table 2

Table 2 - O-Ring Face Seal Cap					
Parker™ Part Number	Tube Outside Diameter (OD) (inch)	Dimension T6 (UN/UNF - 2B)	Dimension B Hex (inch)	Dimension K (inch)	Dimension L (inch)
4 FNL	1/4	9/16-18	11/16	0.34	0.65
6 FNL	3/8	11/16 - 16	13/16	0.39	0.76
8 FNL	1/2	13/16 - 16	15/16	0.45	0.88

Date	Details
2012-9-24	Module Created
2013-10-22	add qsb6.7 and qsl9 cm2350 engine families
2014-3-4	Corrective Action
2014-9-29	Add QSX15 CM2350
2016-4-4	Updated product affected to a generic statement instead of a specific engine list.
2019-6-10	Removed appendix A

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